

Trader Performance vs Bitcoin Market Sentiment

Executive summary

The joined dataset contains **40,872 matched trade rows** from **2023-12-05 to 2025-05-01** across **10 accounts** and **164 coins**. Performance is measured two ways: realized row performance uses nonzero Closed PnL, while overall strategy economics use $\text{net_pnl} = \text{Closed PnL} - \text{Fee}$ across every execution.

Key findings:

- 1. **Best absolute sentiment regime:** Fear generated \$1,721,475 net PnL across 11,523 executions.
- 2. **Lowest net PnL regime:** Extreme Fear still generated positive net PnL at \$353,675, so every sentiment bucket was profitable in this sample.
- 3. **Best capital efficiency:** Extreme Fear produced 20.79 net PnL per \$1k traded.
- 4. **Highest realized win rate:** Neutral had a 91.1% win rate on rows with nonzero Closed PnL.
- 5. Spearman correlation between daily Fear Greed value and daily net PnL is **-0.04**, so the relationship is not simply linear; regime behavior and direction matter more than the raw index level alone.

Performance by sentiment

Sentiment	Trades	Volume	Net PnL	Net PnL / \$1k vol	Realized win rate	Profit factor
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Extreme Fear	1,623	\$17,015,769	\$353,675	20.79	81.2%	4.06
Fear	11,523	\$278,840,188	\$1,721,475	6.17	87.2%	7.57
Neutral	7,620	\$112,155,600	\$660,977	5.89	91.1%	6.27
Greed	11,943	\$197,089,888	\$928,207	4.71	89.6%	4.77
Extreme Greed	8,163	\$73,102,871	\$431,429	5.90	90.8%	3.59

Direction patterns

Long/short behavior changes the story. The direction heatmap shows net PnL per \$1k traded, which is useful because some sentiment regimes have much more volume than others.

Chart reference: *Direction sentiment efficiency*

Charts

Chart reference: Net PnL by sentiment

Chart reference: Rolling net PnL

Strategy implications

- Prefer sizing decisions by sentiment regime instead of treating all days equally. The best absolute PnL regime and the best efficiency regime are the first places to increase review depth.
- Track long and short books separately by sentiment. Direction-level efficiency can reveal whether profits come from being directionally right or simply trading more volume in a regime.
- Use realized win rate together with profit factor. A high win rate with weak profit factor can still be fragile if losing rows are large.
- Add risk controls or lower allocation to the weakest `net_pnl_per_1k_volume` regimes; those periods earn less after normalizing for traded notional.
- Build account-level playbooks: the top account table identifies who performs best in each sentiment regime, which can be converted into trader-specific allocation or copy-trading rules.

Top accounts by regime

Sentiment	Account	Net PnL	Net PnL / \$1k vol	Win rate
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Fear	0x083384f897ee0f19899168e3b1bec365f52a9012	\$1,110,128	36.68	89.1%
Neutral	0x513b8629fe877bb581bf244e326a047b249c4ff1	\$367,920	5.28	96.7%
Fear	0x513b8629fe877bb581bf244e326a047b249c4ff1	\$329,571	1.46	86.0%
Greed	0x083384f897ee0f19899168e3b1bec365f52a9012	\$275,801	31.75	60.1%
Greed	0x430f09841d65beb3f27765503d0f850b8bce7713	\$227,300	193.47	100.0%
Extreme Greed	0x75f7eeb85dc639d5e99c78f95393aa9a5f1170d4	\$210,004	17.26	95.3%
Extreme Fear	0x72c6a4624e1dffa724e6d00d64ceae698af892a0	\$198,844	344.21	49.7%
Fear	0x72c6a4624e1dffa724e6d00d64ceae698af892a0	\$144,452	228.65	91.1%
Greed	0x513b8629fe877bb581bf244e326a047b249c4ff1	\$138,179	1.13	91.3%
Extreme Fear	0x083384f897ee0f19899168e3b1bec365f52a9012	\$124,625	82.69	100.0%
Neutral	0x083384f897ee0f19899168e3b1bec365f52a9012	\$124,349	12.66	100.0%
Extreme Greed	0x430f09841d65beb3f27765503d0f850b8bce7713	\$124,241	999.65	100.0%
Extreme Greed	0x4f93fead39b70a1824f981a54d4e55b278e9f760	\$115,357	2.52	91.7%
Neutral	0x420ab45e0bd8863569a5efbb9c05d91f40624641	\$74,265	366.14	100.0%
Extreme Fear	0x4f93fead39b70a1824f981a54d4e55b278e9f760	\$71,422	7.66	100.0%

Outputs

- `joined_trades_with_sentiment.csv`: original trade rows joined to daily sentiment.
- `sentiment_performance_summary.csv`: regime-level metrics.
- `daily_sentiment_performance.csv`: daily metrics for further charting/modeling.
- `direction_sentiment_summary.csv`: long/short/other breakdown by sentiment.
- `top_coins_by_sentiment.csv`: top five coins by net PnL inside each sentiment regime.
- `top_accounts_by_sentiment.csv`: top three accounts by net PnL inside each sentiment regime.

Notes and assumptions

- Trade dates use Timestamp IST converted to calendar date, then joined to the Fear Greed date.
- Closed PnL is treated as realized PnL at execution-row level. Rows with zero Closed PnL still matter for fee-adjusted economics.
- $\text{net_pnl} = \text{Closed PnL} - \text{Fee}$.
- The supplied historical file does not contain an explicit leverage column, so leverage-specific analysis was not performed.